

UrbanPulse Readiness Checklist

UrbanPulse Readiness Checklist

- Disaster Recovery: Kafka retention plus checkpointed Spark/Flink state support recovery.
- RPO: Stream replay from Kafka topics and retained topic history.
- RTO: Flink and Spark jobs can be restarted from checkpoints and Kafka offsets.
- Data Sovereignty: Topic retention and local deployment keep civic telemetry in controlled infrastructure.
- Open Source: Kafka, Flink, Spark, and Python are open-source components.
- Accessibility: Logging and clear documentation support operator handoff and review.
- Security: TLS and RBAC can be layered over the same topology in production.
- Monitoring: Kafka UI, consumer lag output, and stream job logs provide observability.
- Audit Logging: DLQ records and incident outputs provide auditable evidence trails.
- Backup: Kafka topic retention and Parquet snapshots provide a backup path.
- High Availability: Three Kafka brokers and replicated topics support resilience.
- Role Based Access: Consumer groups and topic-level separation support operational segregation.